

Data Analysis Technical Assessment

Submission Deadline: July 30, 2026 – 11:59 PM

Objective

This is a separate, standalone assessment focused purely on data analysis — no chatbot, no API, no LLM integration required. We want to see how you explore an unfamiliar dataset, reason about a business, and communicate findings clearly.

You'll use the same Drive Finance dataset provided earlier. This task evaluates your analytical thinking, SQL/Pandas fluency, statistical judgment, and ability to turn numbers into a business narrative.

Provided Files

- CSV datasets (same as before): users, employees, finance, loan, installments, commissions, employee_customer_assignment, employee_kpi, territories, hierarchy_levels
- ERD Diagram

If you still have these locally from the previous task, you can reuse them as-is.

Requirements

Part 1 – Exploratory Data Analysis

Produce a clear exploration of the dataset covering:

- Shape and structure of each table, and how they relate (validate against the ERD)
- Data quality: missing values, duplicates, outliers, and any inconsistencies worth flagging — and your judgment on which are real issues vs. expected nulls
- Key distributions: loan amounts, interest rates, tenure, customer credit scores, territory sizes

Part 2 – Business Questions

Answer the following using SQL and/or Pandas. For each answer, show your query/code, the result, and 2–3 sentences of interpretation — a number alone is not an answer.

1. What is the overall loan portfolio health? (active vs. closed vs. delayed, total outstanding balance, average days-past-due)
2. Which territories are over- and under-performing, and by what measure did you decide “performing”?
3. Based on the commission and KPI data available, who are the top and bottom performers, and is target-setting fair across them (same target, different difficulty)? Be clear in your answer about which employees your data actually covers.
4. What does the overdue-installment population look like, based on due dates and payment status? Is there a pattern by territory, employee, loan size, or customer credit score?
5. If you had to flag accounts at risk of default before they become delayed, what signals in this data would you use, and why?

Part 3 – One Original Insight

Find one thing in the data we didn't ask about that you think matters for the business. This is the part we weight most heavily — it's where we see how you think, not just how you execute.

Part 4 – Communication

Package your findings as if presenting to a non-technical business stakeholder:

- A short written summary (1 page max) in plain business language
- Supporting charts (however you like — matplotlib/seaborn/plotly, a BI tool, or even a slide) for your 3–4 most important findings
- Everything reproducible: your code should run end-to-end and regenerate the same charts/numbers

Deliverables

- Source code (notebook or scripts — your choice)
- requirements.txt
- The 1-page summary + charts (PDF, slide, or notebook export — your choice)
- A short note on tools/libraries used and why

Evaluation

Category	Weight
Data Handling & SQL/Pandas Fluency	30%
Analytical Reasoning & Business Judgment	30%
Insight Quality (Part 3)	20%
Communication & Clarity	20%

Notes

- This task does not require an LLM, API, or UI. Keep it focused on the analysis itself.
- We're evaluating judgment more than volume — a few well-reasoned findings beat a long list of shallow ones.
- If anything is ambiguous, state the assumption you made and move on; we'd rather see your reasoning than a clarifying-question thread.

Good luck!